MTH 419A Intro to Abstract Algebra

SAMPLE EXAM 2

Name:

SAMPLE

UB Person Number:

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

Instructions:

- Each problem is worth 25 points.
- You may use course lecture notes and your own notes.
- No textbooks and electronic devices.
- Use backside pages for scratch work only.
- For full credit solve each problem fully, showing all relevant work.

1. Let G be a group and let H be a normal subgroup of G. Show that the quotient group G/H is abelian if and only if for any $a, b \in G$ we have $a^{-1}b^{-1}ab \in H$.

2. a) Let $f: G \to H$ be a homomorphism of groups. Show that if f is 1-1 and $a \in G$ then $|f(a)| = |a|$ (where $|f(a)|$ and $|a|$ denote orders of a and $f(a)$, respectively).

b) Show that there is no 1-1 homomorphism $f: \mathbb{Z}_{25} \to \mathbb{Z}_5 \times \mathbb{Z}_5$.

3. Let G be a group that acts on a set X. Denote by N the set of all elements of G that fix all elements of X:

$$N = \{g \in G \mid gx = x \text{ for all } x \in X\}$$

a) Show that N is a subgroup of G.

b) Show that N is a normal subgroup.

4. a) Find all abelian groups (up to isomorphism) of order 180.

b) For each group from part a) compute the number of elements of order 3 in that group. Justify your answer.

c) Which groups from part a) are cyclic? Justify your answer.